

■ Cash & Bank Management Module & Business Cycles

D365FO Middleware Backend — Cash Receipts, Disbursements & FX Balancing

Target Audience: Treasury, Cash Office, AR/AP Collections, Software Engineers.

1. Business Domain Overview

The **Cash & Bank Management** module handles all treasury and liquidity transactions:

- **Cash-In (Collections & Receipts):** Recording customer payments, bank deposits, down payments, and wire receipts for freight and fleet trucking services.
- **Cash-Out (Disbursements & Advances):** Recording vendor payments, bank transfers, petty cash disbursements, and driver operational cash advances.
- **FX Exchange Rate Balancing:** Adjusting foreign currency receipts/payments against original invoice exchange rates to record realized foreign exchange gains or losses.
- **Custody Settlement Routing:** Isolating custody advance settlements and routing them to dedicated clearing journals.

2. Cash Entry Processors & Business Scope

Processor Class Name	EntryProcessorTypes	D365FO Target Entity	Real-World Business Context
CashInFreightEntryProcessor	CashInFreight	CustPaymentJournalHeaders / Lines	Customer Payment Receipts (Freight)
CashInTruckingEntryProcessor	CashInTrucking	CustPaymentJournalHeaders / Lines	Customer Payment Receipts (Trucking)
CashOutFreightEntryProcessor	CashOutFreight	Disbursement Journal	Vendor & Cash Disbursements (Freight)
CashOutTruckingEntryProcessor	CashOutTrucking	Disbursement Journal	Vendor & Cash Disbursements (Trucking)

3. Financial & Accounting Impact in D365FO

■ Cash-In (Customer Collection Journal - Cust - Pay):

- **DEBIT:** Bank / Cash Main Account (e.g., **110201** CIB EGP Bank or **110202** USD Bank)
- **CREDIT:** Customer Subledger Account (**CustAccount** e.g. **CUST-00192**)

- **DEBIT/CREDIT (FX Difference):** Realized FX Gain/Loss Account (e.g., **710101** / **810101**) if invoice FX rate differs from payment date FX rate.

■ **Cash-Out (Vendor Disbursement Journal):**

- **DEBIT:** Vendor Subledger Account (**VendAccount** e.g. **VEND-00412**) or Expense/Custody Account
- **CREDIT:** Bank / Cash Main Account

4. End-to-End Cash Business Cycle Flow

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[Bank Statements / Cash Office Receipt Vouchers]
■
▼
[API Endpoint: Upload Cash Receipts Excel]
■
▼
[CashInFreightEntryProcessor]
(Parses Bank Accounts, Customer IDs, FX Differences)
■
▼
[DataBatch Status: PENDING (Max 1000 Lines)]
■
▼
[BullMQ Queue: dfo-customer-payment-journal-queue]
■
▼
[CustomerPaymentPostingStrategy]
■
▼
[Post OData to D365FO CustPaymentJournalHeaders]
■
▼
[D365FO Posts Customer Payment Journal]
(Clears Open Invoice & Balances Bank/Customer Subledgers)
■
▼
[Batch Status Set to COMPLETED]

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Detailed Step-by-Step Cycle Execution:

1. **Cash Sheet Upload:** Cashier or treasury accountant uploads daily collection logs containing bank account codes, customer codes, payment amounts, and invoice reference numbers.
2. **Custody Split:** Any lines marked as driver custody settlements are separated and routed to the custody settlement workflow.
3. **FX Calculation:** If a customer pays in USD for an invoice originally issued in USD but recorded in EGP, the system fetches transaction date exchange rates to balance net GL amounts.
4. **Queue Posting:** Dispatched to **dfo-customer-payment-journal-queue**.
5. **OData Execution:** Posts header and line records into D365FO customer payment journal.
6. **Subledger Settlement:** D365FO applies the cash receipt against open customer invoices, updating customer balance.

5. Detailed Cash-In (Collections) Deep Dive

The **Cash-In** process manages the intake of customer collections. It translates operational receipts (cash, cheques, POS, bank transfers) into D365FO Customer Payment Journals (**Cust - Pay**).

5.1 The Business Lifecycle

1. **Receipt Generation:** A customer pays for a freight or trucking service. The cashier issues a receipt (Voucher) specifying the payment method, amount, currency, and the invoice being settled.
2. **Batch Upload:** At the end of the day or shift, the treasury team uploads an Excel sheet containing all receipts.
3. **Identification & Validation:** The system verifies the Customer Account against D365FO master data and ensures the referenced Invoice exists and is posted in D365FO.
4. **FX Gain/Loss Evaluation:** If the customer pays today for an invoice issued last month, the exchange rates likely differ. The system calculates the realized exchange gain or loss.
5. **Notes Receivable:** If the payment is made via post-dated cheques, it is classified under specific Notes Receivable ledger accounts instead of immediate cash bank accounts.
6. **ERP Settlement:** D365FO receives the data, debits the bank/cash account, credits the customer account, and settles (closes) the open invoice.

5.2 The Code & Logic Lifecycle

1. **Data Ingestion & SafeType Parsing:**
 - The **CashInFreightEntryProcessor** processes rows.
 - It filters by **SafeType**. If **SafeType** is **Custody Settlement**, the line is extracted and redirected to the Custody Settlement logic. Only **Customer Collection** or **DownPayment** proceeds in the main Cash-In flow.
2. **Grouping by Invoice (Uniqueld):**
 - Lines are grouped by a generated **UniqueId** representing the invoice group. Typically, a group consists of a debit line (Bank) and a credit line (Customer).
3. **Invoice Format Normalization:**
 - The **INVOICE** column is normalized to match D365FO's strict 9-digit format + suffix rule (e.g., **000012345/INVOICE** for import, **000012345/OF - FW** for export freight).
 - The processor validates that this invoice number exists and is fully posted in D365FO via a pre-validation query.
4. **Exchange Rate (FX) Calculation:**
 - The processor fetches the **EXCHANGERATESECONDARY** or the daily rate from the D365FO **ExchangeRates** endpoint.
 - It calculates the difference between the invoice's original EGP value and the payment's current EGP value.
 - Any difference creates an automated balancing line pointing to the FX Gain (**710101**) or FX Loss (**810101**) Main Account to ensure **Sum(Debit) == Sum(Credit)**.
5. **Notes Receivable Handling:**
 - If the offset line uses Ledger account types like **122201**, **122202**, etc., the system overrides the **OffsetAccountType** to **Bank** and sets the **PaymentMethod** to **NR** (Notes Receivable).
6. **Payload Construction:**
 - Maps the enriched group into **CustPaymentJournalHeaders** (the batch envelope) and **CustPaymentJournalLines** (the individual debits/credits).

6. Detailed Cash-Out (Disbursements) Deep Dive

The **Cash-Out** process governs the outflow of funds from the company to vendors, subcontractors, or employees (custody).

6.1 The Business Lifecycle

1. **Payment Requisition:** A vendor payment is approved, or an operational coordinator requests a cash advance (custody) for a trip's port fees.
2. **Disbursement:** The cashier releases the funds via bank transfer, cheque, or petty cash. The transaction is logged in the daily disbursement sheet.
3. **Tax & Withholding Deduction (WHT):** Before paying a vendor, Egyptian tax law often requires deducting a percentage (e.g., 1% or 3%) as Withholding Tax. The payment reflects the net amount.
4. **FX Adjustments:** If paying a foreign shipping line in USD, the rate paid today might differ from the rate when the vendor invoice was booked, resulting in an FX gain/loss.
5. **ERP Settlement:** D365FO records the disbursement, crediting the Bank/Cash account and debiting the Vendor Subledger (or the employee Custody account).

6.2 The Code & Logic Lifecycle

1. **Data Ingestion & SafeType Parsing:**
 - The **CashOutFreightEntryProcessor** processes rows.
 - It checks **SafeType** to distinguish between **Vendor Payment**, **Custody Issue**, and **Direct Expense**.
2. **Vendor Lookup & Validation:**
 - The **Vendor** dimension is extracted. The processor queries the Master Data cache for the vendor profile to ensure it is active and not blocked for payment.
3. **Withholding Tax (WHT) Automation:**
 - The processor checks the vendor's tax profile and the Excel **ITEMWITHHOLDINGTAXGROUPCODE** column.
 - If WHT applies, the processor automatically calculates the tax amount, reduces the vendor's net payable line, and adds a credit line to the WHT Payable Main Account.
4. **Exchange Rate (FX) Calculation:**
 - Similar to Cash-In, the processor fetches the daily exchange rate for the **TRANSDATE**.
 - It compares it against the original vendor bill rate, generating automatic FX Gain/Loss ledger lines to keep the voucher perfectly balanced.
5. **Payload Construction:**
 - The data is transformed into **Ledger JournalHeaders** (configured for AP Disbursement journals) and **Ledger JournalLines**.
 - Lines with **ACCOUNTTYPE = 'Vend'** hit the subledger, while **ACCOUNTTYPE = 'Bank'** record the outflow of cash.